to those of Eq. 33-13, with only two minor changes. The value of *k* is different (as given in Eq. 33-14), and two coefficients, a_1 and b_1, are negated in value. These equations are carried out in lines 1330 to 1410 in the program, providing the desired cutoff frequency, and the choice of a high-pass or low-pass response.

The Best and Worst of DSP

This book is based on a simple premise: **most DSP techniques can be used and understood with a minimum of mathematics**. The idea is to provide scientists and engineers *tools* for solving the DSP problems that arise in their non-DSP research or design activities.

These last four chapters are the other side of the coin: DSP techniques that can *only* be understood through extensive math. For example, consider the Chebyshev-Butterworth filter just described. This is the *best* of DSP, a series of elegant mathematical steps leading to an optimal solution. However, it is also the *worst* of DSP, a design method so complicated that most scientists and engineers will look for another alternative.

Where do you fit into this scheme? This depends on *who* your are and *what* you plan on using DSP for. The material in the last four chapters provides the theoretical basis for signal processing. If you plan on pursuing a *career* in DSP, you need to have a detailed understanding of this mathematics. On the other hand, specialists in other areas of science and engineering only need to know how DSP is *used*, not how it is *derived*. To this group, the theoretical material is more of a background, rather than a central topic.